

Problem set #3

MATH 4503/6503: Numerical Methods for Differential Equations

Due date: March 9, 2026

Reading resources: Notes uploaded to D2L; book pages 137-166

Problem #1: filling in the gap ($M = 3$)

Hey, guys! Apologies for uploading this problem set this late! Anyway, let us begin. We have covered $M = 2$ and $M = 4$ Runge-Kutta stencils in quite some detail in class, so I felt like using this problem to fill in the gap: $M = 3$.

- (a) First, I would want you to find the $\alpha_{n,m}, b_n, c_n$ Runge-Kutta coefficients for the $M = 3$ method using the same techniques we used for the $M = 2, 4$ methods. (*Hint: After setting a suitable portion of your Butcher tableau equal to 0 for an explicit stencil, use the assumptions $c_2 = 1/2$ and $c_3 = 3/4$.*)
- (b) Now, use the stencil you have constructed in part (a) to calculate $y(1)$ for the initial value problem

$$y'(x) = -y(x), y(0) = 1.$$

Also perform an error analysis to confirm that you get the correct global error at $x = 1$.

Problem #2: damped harmonic oscillator

We will make a light start on stability here. As you might already know, a harmonic oscillator is a mass attached to a spring; a damped harmonic oscillator just has the mass experiencing some friction to its motion. (Imagine a wooden box attached to a spring and made to move on a rough horizontal table.) The second order ODE representing its motion is

$$\ddot{u}(t) + 2\Gamma\dot{u}(t) + \omega^2 u(t) = 0, \quad (*)$$

where Γ and ω are nonnegative constants (with Γ capturing the amount of friction) and overdots denote time derivatives. You might want to load the `LinearAlgebra` package for this problem as it will be useful. Recall that the command for this is `with(LinearAlgebra)`.

- (a) Using $v(t) \equiv \dot{u}(t)$, convert $(*)$ above into a coupled set of ODEs for $u(t)$ and $v(t)$.

- (b) This set of ODEs can be rewritten as the matrix equation

$$\dot{\mathbf{y}}(t) = \mathbf{A}\mathbf{y}(t), \mathbf{y}(t) = \begin{bmatrix} u(t) \\ v(t) \end{bmatrix}. \quad (**)$$

Look up the help pages for `GenerateMatrix`, especially the last example. Use `GenerateMatrix` to find the 2×2 matrix $\mathbf{A}$.

- (c) The `Eigenvalues(A)` command will return a vector containing the eigenvalues of $\mathbf{A}$ and a matrix $\mathbf{\Lambda}$ whose columns are the eigenvectors of $\mathbf{A}$. Define a new set of variables $\mathbf{x}(t)$ by the relation $\mathbf{y}(t) = \mathbf{\Lambda}\mathbf{x}(t)$. Hence, show that $\mathbf{x}(t)$ satisfies

$$\dot{\mathbf{x}}(t) = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \mathbf{x}(t),$$

where the λ_n with $n = 1, 2$ are the eigenvalues of $\mathbf{A}$. You might ask what we achieve through this exercise. Well, we have demonstrated that solving $(**)$ is equivalent to solving two decoupled ODEs, namely $x_1 = \lambda_1 x_1$ and $x_2 = \lambda_2 x_2$. (For clarity, $x_1(t)$ and $x_2(t)$ are the two components of the vector $\mathbf{x}(t)$.) This suggests that if a particular numerical stencil is stable for these two scalar problems, it would also be stable for the matrix problem $(**)$. So, based on the eigenvalues of $\mathbf{A}$, do you expect the forward Euler scheme to be stable for the given problem when $\Gamma = 0$?

- (d) To test your intuition in part (c), consider now the numerical solution of $(**)$ directly. We assume initial data $\mathbf{y}(0) = \mathbf{y}_0$ and step size h . Adapting forward Euler and trapezoidal schemes to the problem at hand, we obtain the generalizations

$$\mathbf{y}_{i+1} = \mathbf{y}_i + h\mathbf{A}\mathbf{y}_i, \mathbf{y}_{i+1} = \mathbf{y}_i + \frac{1}{2}h(\mathbf{A}\mathbf{y}_i + \mathbf{A}\mathbf{y}_{i+1}), \quad (***)$$

respectively. As usual, $\mathbf{y}_i$ denotes the numeric approximation to the true solution at $t = t_i$; that is, $\mathbf{y}_i \approx \mathbf{y}(t_i)$. Each stencil can be rewritten as

$$\mathbf{y}_{i+1} = \mathbf{P}\mathbf{y}_i.$$

Find the matrix $\mathbf{P}$ for each stencil. (Note that the product of two matrices in Maple is `A.B` and the inverse of a matrix is `A^(-1)`.)

- (e) Let us now make the following assumptions:

$$\omega = 1, \Gamma = 0, \mathbf{y}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Write two Maple procedures to solve $(**)$ numerically for $t \in [0, t_0]$ using the stencils $(***)$. The arguments of your procedures should include the final time t_0 and the number of steps N . From your results, is each stencil stable or unstable for this problem? (*Hint*: See the notes on absolute stability, and perform your numerical experiments along the same lines as we did in class.)

- (f) *Mandatory for graduate students & bonus for undergraduates*: Here is some extra work

for the grad students. It is more like drudgery since you have done a similar problem on the previous problem set; however, I am choosing to give it anyway since the undergrads too have been through a similar exercise, so this would possibly give them a nice chance to score a bonus!

Anyway, what I want you to do here is to note that the one-step error for each stencil is defined by

$$\mathbf{P}\mathbf{y}(t) - \mathbf{y}(t+h) = O(h^{p+1}),$$

where p is a positive integer. Find this p for each stencil. Next, confirm that the global error is $O(h^p)$ for each stencil by comparing the numerical and analytic values for the norm of the solution $\|\mathbf{y}(t)\| = \mathbf{y}^T \mathbf{y}$ evaluated at $t = 50$ when $\omega = 1$, $\Gamma = 1/2$, and $\mathbf{y}_0 = \begin{bmatrix} 1 & 0 \end{bmatrix}^T$.

Problem #3: relative stability

Our in-class discussion revolved around absolute stability entirely. We saw that a number of stencils reduce down to $y_{i+1} = P(z)y_i$ when applied to $y'(x) = \lambda y(x)$, where $z = h\lambda$. To leading order, the numerical error obeys $E_{i+1} = P(z)E_i$ for such stencils. If we wish to avoid exponentially growing errors, we need to enforce the absolute stability criterion $|P(z)| \leq 1$.

However, the true solution to this problem, $y(x) = y(0)e^{\lambda x}$, is itself exponentially growing or decaying. One might argue that if $y(x)$ is itself decaying, then it isn't good enough to simply have the past errors decaying too—they should be decaying at a faster rate. Conversely, if the true solution is growing exponentially, then perhaps it is fine for the errors also to be growing as long as they are not growing faster. This motivates us to call a stencil relatively stable if the following condition is met:

$$\left| \frac{E_{i+1}}{E_i} \right| \leq \left| \frac{y(x+h)}{y(x)} \right|.$$

On the right-hand side, $y(x)$ and $y(x+h)$ are values of the analytic solution of $y'(x) = \lambda y(x)$. This definition means that a stencil will be relatively stable as long as the numerical errors are growing slower than the actual solution. Use the same techniques as we used in class to plot absolute stability regions to plot regions of relative stability in the complex z plane for the forward and backward Euler methods, the trapezoidal method, the Heun method, and the fourth order Runge-Kutta method.

Problem #4: convergence without analytic solution

For obvious reasons, we most often employ numerical methods to solve differential equations for which the analytic solution is not known. For example, using `dsolve` on the problem

$$\frac{dy}{dx} = \frac{1}{1 + \sin(y)}, y(0) = -\frac{\pi}{4}$$

in Maple results in a transcendental equation for y with no closed form solution. In these cases, it is still possible to test the rate of convergence of a numerical solution to the true solution even though the true solution is not known. Suppose we have a stencil with one-step error $O(h^{p+1})$ and we use this stencil to solve the above initial value problem on the interval $x \in [0, x_0]$ twice,

once with a step size $2h$ and once with h . The numeric solution generated when the step size is $2h$ will be denoted by $y_i^{(2h)} \approx y(2hi)$ with $i = 0, 1, 2, \dots, N$ (remember $y(x)$ is the true solution). On the other hand, when the step size is h , the numeric solution is labeled as $y_i^{(h)} \approx y(hi)$ with $i = 0, 1, 2, \dots, 2N$. Here, $h = x_0/2N$. Consider the quantity

$$\xi(h, x_0) = \sqrt{\frac{1}{N+1} \sum_{i=0}^N \left(y_i^{(2h)} - y_{2i}^{(h)} \right)^2}.$$

We call $\xi(h, x_0)$ the norm between the numeric solutions with step sizes $2h$ and h .

- (a) If the numeric method we are using to calculate the above norm is zero-stable, we would expect $\xi(h, x_0) = O(h^q)$ in the $h \rightarrow 0$ limit. What do you think is the value of q in terms of p ? (*Hint*: Express both $y_i^{(2h)}$ and $y_{2i}^{(h)}$ as " $y(2hi) + \text{global error}$ " first.)
- (b) Write a two Maple procedures to calculate $\xi(h, x_0)$ using numeric output from the Heun and fourth order Runge-Kutta methods as applied to specific initial value problem above. Remember to define N using the `round` command.
- (c) Plot $\xi(h, x_0)$ versus h on a log-log plot for $x_0 = 2, 4, 6$ and for both stencils. Does the small h behavior match your expectation from part (a)?